- Outer conductor radius = b
- Charge per unit length on the inner conductor = $+\lambda$
- Outer conductor carries an equal and opposite charge $-\lambda$

Gauss's Law:

For a cylindrical surface of radius r such that $a < r < b$:

$$\oint \vec{E} \cdot d\vec{A} = \frac{q_{\text{enc}}}{\varepsilon_0}$$

$$E(2\pi r L) = \frac{\lambda L}{\varepsilon_0}$$

$$E(r) = \frac{\lambda}{2\pi\varepsilon_0 r}$$

Where:

- ε_0 is the vacuum permittivity
- $E(r)$ is the radial electric field at radius r

This shows the electric field exists only in the dielectric region between $a < r < b$, and falls off as $1/r$.

Potential Difference Between Conductors

$$V = -\int_a^b E(r)\,dr = -\int_a^b \frac{\lambda}{2\pi\varepsilon_0 r}\,dr = \frac{\lambda}{2\pi\varepsilon_0}\ln\left(\frac{b}{a}\right)$$

Capacitance per Unit Length

Ask anything

Tools

ChatGPT can make mistakes. Check important info.